# Self-Check Questions: Power Series and Coefficient Matching

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1. What does it mean for two power series to be equal?

   *Answer.* Two power series $\sum_{n=0}^{\infty} a_n x^n$ and $\sum_{n=0}^{\infty} b_n x^n$ are equal if they converge to the same value for every $x$ in some open interval $(-R, R)$ with $R > 0$. By the uniqueness theorem this forces $a_n = b_n$ for all $n \geq 0$.

2. Under what conditions can two power series be compared coefficient-by-coefficient?

   *Answer.* Both series must (i) converge on a common open interval containing 0 and (ii) represent the same function on that interval. Under these conditions the uniqueness theorem guarantees that corresponding coefficients are equal.

3. Why is the statement

   $$\sum_{n=0}^{\infty} c_n x^n = 0$$

   stronger than merely saying that the series equals zero at $x = 0$?

   *Answer.* The equation at $x = 0$ alone yields only $c_0 = 0$. The identity as a power series asserts that the sum is zero *for every* $x$ in some interval around 0, which (by the uniqueness theorem) forces every coefficient $c_n$ to vanish.

4. State the uniqueness theorem for power series.

   *Answer.* If $\displaystyle\sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} b_n x^n$ for all $x$ in some open interval containing 0, then $a_n = b_n$ for all $n \geq 0$.

5. How can the uniqueness theorem be proved using derivatives?

   *Answer.* Let $f(x) = \sum_{n=0}^{\infty} c_n x^n = 0$ on $(-R, R)$. Setting $x = 0$ gives $c_0 = 0$. A convergent power series may be differentiated term-by-term inside its interval of convergence, so $f'(x) = \sum_{n=1}^{\infty} n c_n x^{n-1} = 0$. Setting $x = 0$ gives $c_1 = 0$. Repeating, the $k$-th derivative satisfies $f^{(k)}(0) = k!\, c_k = 0$, hence $c_k = 0$ for every $k \geq 0$.

6. Why does

   $$\sum_{n=0}^{\infty} c_n x^n = 0$$

   imply $c_0 = 0$?

   *Answer.* Substitute $x = 0$. Every term with $n \geq 1$ contains a factor of $x$ and therefore vanishes, leaving $c_0 \cdot 1 = 0$, so $c_0 = 0$.

7. Why does differentiating the series imply $c_1 = 0$?

   *Answer.* Differentiating term-by-term gives $\sum_{n=1}^{\infty} nc_n x^{n-1} = 0$. Setting $x = 0$ in this new series yields $1 \cdot c_1 = 0$, so $c_1 = 0$.

8. What is the general formula for $c_n$ in terms of derivatives of the sum at $x = 0$?

   *Answer.* If $f(x) = \sum_{n=0}^{\infty} c_n x^n$, then differentiating $n$ times and evaluating at $x = 0$ gives $f^{(n)}(0) = n!\, c_n$, so

$$c_n = \frac{f^{(n)}(0)}{n!}.$$

9. Rewrite

$$\sum_{n=1}^{\infty} a_n x^{n-1}$$

   as a power series in $x^n$.

   *Answer.* Substitute $m = n - 1$ (so $n = m + 1$); when $n = 1$, $m = 0$:

$$\sum_{n=1}^{\infty} a_n x^{n-1} = \sum_{m=0}^{\infty} a_{m+1}\, x^m.$$

   Relabelling $m$ back to $n$: $\displaystyle\sum_{n=0}^{\infty} a_{n+1}\, x^n$.

10. Why is reindexing necessary before matching coefficients?

    *Answer.* Coefficient matching works by equating the multipliers of the *same* power $x^n$ on both sides. If the series are expressed in different powers (e.g. $x^{n-1}$ versus $x^n$) the $k$-th summand of each series corresponds to a different power of $x$, so naive comparison produces a wrong (or meaningless) relation.

11. What mistake occurs if two sums involve different powers of $x$ and you compare coefficients immediately?

    *Answer.* You would equate the coefficient of $x^n$ in one series against the coefficient of $x^n$ in the other, but one series might have its $n$-th term equal to $x^{n-1}$ rather than $x^n$. The resulting "recurrence" would relate coefficients that actually belong to different powers of $x$, giving an incorrect equation.

12. Given

$$\sum_{n=0}^{\infty} a_n x^n + \sum_{n=2}^{\infty} b_n x^n,$$

    what is the coefficient of $x^1$?

    *Answer.* The first series contributes $a_1$ for the power $x^1$. The second series starts at $n = 2$, so it has no $x^1$ term. The coefficient of $x^1$ is $a_1$.

13. What is the coefficient of $x^5$?

    *Answer.* Both series include $n = 5$, so the coefficient of $x^5$ is $a_5 + b_5$.

14. Suppose

$$\sum_{n=0}^{\infty} a_n x^n + \sum_{n=3}^{\infty} b_n x^n = 0.$$

Why must $a_0, a_1, a_2$ be treated separately?

*Answer.* The second series has no terms for $n = 0, 1, 2$. For those powers of $x$, the only contribution to the combined sum comes from the first series alone, so the equation $(\cdots) = 0$ for each of $x^0$, $x^1$, $x^2$ involves only $a_0$, $a_1$, $a_2$ respectively, and each must vanish on its own.

15. What equation do you obtain from the coefficient of $x^0$?

*Answer.* $a_0 = 0$.

16. What equation do you obtain from the coefficient of $x^1$?

*Answer.* $a_1 = 0$.

17. What equation do you obtain from the coefficient of $x^2$?

*Answer.* $a_2 = 0$.

18. What equation do you obtain for $n \geq 3$?

*Answer.* Both series contribute, so $a_n + b_n = 0$, i.e. $b_n = -a_n$ for all $n \geq 3$.

19. How would the answer change if the second series started at $n = 5$?

*Answer.* Then $a_0, a_1, a_2, a_3, a_4$ would each be forced to zero individually (isolated terms), and for $n \geq 5$ one would obtain $a_n + b_n = 0$.

20. If

$$y = \sum_{n=0}^{\infty} a_n x^n,$$

what is $y'$?

*Answer.*

$$y' = \sum_{n=1}^{\infty} n\, a_n\, x^{n-1}.$$

21. What is $y''$?

*Answer.*

$$y'' = \sum_{n=2}^{\infty} n(n-1)\, a_n\, x^{n-2}.$$

22. Why do the sums for $y$, $y'$, and $y''$ begin at different indices?

*Answer.* Differentiating a constant term produces zero, so it drops out of the sum. The constant term $(n = 0)$ of $y$ vanishes in $y'$, forcing $y'$ to start at $n = 1$. The $n = 0$ and $n = 1$ terms of $y'$ collectively contribute at most a constant, which vanishes in $y''$, so $y''$ starts at $n = 2$.

23. After differentiating, how do you reindex so that all series are expressed in powers of $x^n$?

*Answer.* For $y'$, substitute $m = n - 1$ ($n = m + 1$, lower limit $m = 0$):

$$y' = \sum_{m=0}^{\infty} (m+1)\, a_{m+1}\, x^m.$$

For $y''$, substitute $m = n - 2$ ($n = m + 2$, lower limit $m = 0$):

$$y'' = \sum_{m=0}^{\infty} (m+2)(m+1)\, a_{m+2}\, x^m.$$

Rename $m \to n$ in both to obtain all three series in powers of $x^n$ starting at $n = 0$.

24. Why is matching coefficients impossible until all sums use the same power $x^n$?

*Answer.* The uniqueness theorem equates multipliers of identical powers of $x$. If one series runs in $x^n$ and another in $x^{n-2}$, the $k$-th term of each represents a different power of $x$, so there is no consistent meaning to "the coefficient of $x^n$" across the two sums until they are expressed uniformly.

25. When solving an ODE by power series, what usually produces the recurrence relation?

*Answer.* After substituting the ansatz $y = \sum a_n x^n$, reindexing all series so they run in the same power $x^n$, and combining into a single sum equal to zero, equating the coefficient of $x^n$ to zero yields the recurrence relation, which expresses $a_{n+k}$ (for some fixed $k$) in terms of earlier coefficients.

26. What usually determines the first few coefficients separately from the recurrence relation?

*Answer.* The initial conditions of the ODE — typically $y(0)$ and $y'(0)$ for a second-order equation — fix $a_0$ and $a_1$ directly. The recurrence relation then determines all subsequent coefficients in terms of these two. Additionally, isolated low-index terms (powers of $x$ not covered by the combined recurrence) must be set to zero individually, sometimes also pinning early coefficients.

27. Why is coefficient matching essentially equivalent to evaluating derivatives at $x = 0$?

*Answer.* The coefficient $c_n$ of a power series satisfies $c_n = f^{(n)}(0)/n!$. Requiring $\sum a_n x^n = \sum b_n x^n$ and then matching $a_n = b_n$ is therefore exactly the same as requiring $f^{(n)}(0) = g^{(n)}(0)$ for all $n \geq 0$, i.e. that all derivatives of the two functions agree at 0.

28. Could coefficient matching fail for a Taylor series outside its radius of convergence?

*Answer.* Yes. A Taylor series converges to the original function only inside the radius of convergence $R$. Outside $|x| < R$ the series may diverge or converge to a different value, so the identity theorem (which underpins coefficient matching) no longer applies, and matching coefficients yields no information about the function's behaviour there.

29. Why is analyticity essential for the uniqueness theorem?

*Answer.* A smooth ($C^\infty$) function need not equal its Taylor series anywhere except at the centre. If $f$ is not analytic, two distinct functions can share every Taylor coefficient at 0 and still differ on every neighbourhood of 0. Analyticity is precisely the condition that $f$ *equals* its Taylor series in some disk, which is what makes the map $f \mapsto (c_n)$ injective and coefficient matching valid.

30. Give an example of a smooth ($C^\infty$) function whose Taylor series is identically zero but whose function is not.

    *Answer.* Define
    $$f(x) = \begin{cases} e^{-1/x^2} & x \neq 0, \\ 0 & x = 0. \end{cases}$$
    Every derivative of $f$ at $x = 0$ equals 0, so the Taylor series of $f$ centred at 0 is $\sum 0 \cdot x^n \equiv 0$, yet $f(x) > 0$ for all $x \neq 0$.

31. Why does such an example show that power series methods are fundamentally about analytic functions, not merely differentiable functions?

    *Answer.* This function is $C^\infty$ but its Taylor series does not converge to it (the series is identically 0 while the function is not). Any method that represents a solution as a power series and then extracts the solution by matching or summing those series is implicitly assuming the solution is analytic. For merely smooth functions the power series may be useless as a representation, and the method breaks down entirely.

32. In a power series solution of an ODE, what role do the arbitrary constants $a_0$ and $a_1$ typically play for a second-order equation?

    *Answer.* They correspond to the two constants of integration in the general solution. Concretely, $a_0 = y(0)$ and $a_1 = y'(0)$. Every higher coefficient $a_n$ ($n \geq 2$) is determined by the recurrence relation in terms of $a_0$ and $a_1$, so the general power series solution is a linear combination of two independent series, one proportional to $a_0$ and the other to $a_1$.

33. Why do second-order ODEs usually leave two coefficients free?

    *Answer.* By the existence-uniqueness theorem for linear ODEs, the solution space of a homogeneous second-order linear ODE is two-dimensional. In the power series approach this is reflected by the recurrence relation: it expresses $a_{n+2}$ in terms of $a_n$ and possibly $a_{n+1}$, advancing the index by 2 (or by 1), so the two "seed" values $a_0$ and $a_1$ propagate into the entire sequence and remain free parameters.

34. How would you explain, in one paragraph, why "equating coefficients" is mathematically justified rather than a formal trick?

    *Answer.* Equating coefficients is justified by the uniqueness theorem for analytic functions: if two power series converge to the same function on any open interval containing 0, their coefficients must be identical. When we write an ODE as a single power series equal to zero, we are asserting that the resulting function is identically zero on some interval — not just at one point. The uniqueness theorem then guarantees that each individual coefficient must be zero, so we are not merely "pretending" that powers of $x$ are independent; we are using the proved injectivity of the map from an analytic function to its Taylor-coefficient sequence. The method is therefore as rigorous as any other that invokes this theorem, provided the solution is known (or assumed) to be analytic in a neighbourhood of the expansion point.